

Linear Algebra – Problem Set 3 — Solutions

Linear Independence, Bases & Dimension · Full solutions · Original problems in the spirit of MIT 18.06 (Strang)

Problem 1

In $\mathbb{R}^3$, decide whether the three vectors are linearly independent, and justify it by elimination rather than by eye:

Put the vectors in as columns and clear below the first pivot:

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - R_1 : \quad \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & -2 \end{bmatrix}, \quad \text{then } R_3 \leftarrow R_3 - 2R_2 : \quad \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

There is a zero row, so the system $Ac = 0$ has free variables: $\text{rank} = 2 < 3$, and the vectors are **dependent**.

The determinant agrees: expanding along the second row, $\det[u_1 \ u_2 \ u_3] = 0$. (Everything is zero there.)

The relation: the second column is twice the first, so $u_2 - 2u_1 = 0$ is a nontrivial relation with coefficients $(c_1, c_2, c_3) = (-2, 1, 0)$.

$\text{rank} = 2$, $\det = 0$, **dependent**; $u_2 = 2u_1$.

Problem 2

In $\mathbb{R}^3$ consider

(a) Reduce the matrix with all four columns:

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 0 & 1 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix},$$
$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Three pivots, so the span is all of $\mathbb{R}^3$ and its dimension is 3. A spanning set of 4 vectors with a redundancy must therefore lose at least one; any 3 that survive independent will do. Take

$$\mathcal{B} = \{u_1, u_2, u_4\}. \quad \text{Check: } \det[u_1 \ u_2 \ u_4] = \begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 4 \\ 0 & 1 & 1 \end{vmatrix} = 1 \neq 0.$$

The determinant is 1, so those three are independent, and three independent vectors in $\mathbb{R}^3$ already span everything: **no set of two can span $\mathbb{R}^3$** (that is Problem 11).

- (b) Solve $\mathcal{B}c = v$ for each vector. The three basis vectors have trivial coordinates, so the only real work is u_3 : solve $au_1 + bu_2 + cu_4 = u_3$, i.e.

$$\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 4 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \implies a + 2c = 1, 2a + b + 4c = 3, b + c = 1.$$

From the first and third, $a = 1 - 2c$ and $b = 1 - c$. Substituting into the middle equation:

$$2(1 - 2c) + (1 - c) + 4c = 3 \implies 3 - c = 3 \implies c = 0, \quad a = 1, \quad b = 1.$$

$$u_3 = 1u_1 + 1u_2 + 0u_4 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \quad \checkmark$$

So the full coordinate list in the basis $\mathcal{B} = \{u_1, u_2, u_4\}$ is

$$[u_1] = (1, 0, 0)^T, \quad [u_2] = (0, 1, 0)^T, \quad [u_3] = (1, 1, 0)^T, \quad [u_4] = (0, 0, 1)^T.$$

- (c) The dropped vector is u_3 , and the relation is $u_3 - u_1 - u_2 = 0$. Had we tried to keep u_1, u_2, u_3 and drop u_4 instead, we would have found *no* solution: those three span a plane (the same relation kills the third dimension), and u_4 is not in it — check the first coordinate of $\alpha u_1 + \beta u_2 + \gamma u_3 = u_4$: $\alpha + \gamma = 2, 2\alpha + \beta + 3\gamma = 4, \beta + \gamma = 1$ are inconsistent.

A basis is $\{u_1, u_2, u_4\}$; the coordinates are $(1, 0, 0), (0, 1, 0), (1, 1, 0), (0, 0, 1)$ in that order, and $u_3 = u_1 + u_2$ is the vector that drops out.

Problem 3

True or false? Give a counterexample or a correction for each false one.

- (a) **True.** Four vectors in $\mathbb{R}^3$ put into a 3×4 matrix A give $Ac = 0$ with 4 unknowns and 3 equations — a free variable is guaranteed, so a nontrivial solution exists.
- (b) **False.** Any set containing 0 is dependent: the equation $1 \cdot 0 = 0$ is already a nontrivial relation (the coefficient of 0 is nonzero, and the rest are 0).
- (c) **True.** $av + bw = 0$ with $b \neq 0$ gives $w = -\frac{a}{b}v$; conversely if $w = cv$ then $w - cv = 0$ is nontrivial when $v \neq 0$. (The nonzero proviso matters: $v = 0$ is a scalar multiple of everything *and* makes every set dependent.)
- (d) **True.** A relation among three of them would be a relation among all four, so it would extend by zero coefficients to the full set.
- (e) **False.** Adding can only work until the space runs out. In $\mathbb{R}^3$, add $e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ to the independent set $\{e_1, e_2\}$ and you may add that one; adding $e_1 + e_2$ as well gives four vectors in $\mathbb{R}^3$, which by (a) must be dependent — indeed e_3 is not involved: $(e_1 + e_2) - e_1 - e_2 = 0$ is already a relation.

(a) true, (b) false, (c) true, (d) true, (e) false.

Problem 4

Let $u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $w = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$ and $z = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

(a) $ac_1 + bc_2 = w$ is the two-unknown system

$$\begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}. \quad \text{Take } c_1 = 1, c_2 = 1 : \quad 1 + 2 = 3, 2 + 0 = 2, 3 + 1 = 4 \checkmark$$

(b) Same system with right side z :

$$\begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \xrightarrow{\text{row reduce}} \text{inconsistent (a row of zeros} = \text{a nonzero entry)}$$

Explicitly, the equations are $c_1 + 2c_2 = 0$, $2c_1 = 1$, $3c_1 + c_2 = 0$. The second gives $c_1 = \frac{1}{2}$, so the first gives $c_2 = -\frac{1}{4}$; but then $3c_1 + c_2 = \frac{3}{2} - \frac{1}{4} = \frac{5}{4} \neq 0$. Contradiction.

(c)

$$\det[u \ v \ z] = \begin{vmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 3 & 1 & 0 \end{vmatrix} = 1 \cdot (0 \cdot 0 - 1 \cdot 1) - 2 \cdot (2 \cdot 0 - 1 \cdot 3) + 0 = -1 + 6 = 5.$$

Nonzero, so the three are independent.

(a) yes, $w = u + v$; **(b) no**, $z \notin \text{span}(u, v)$; **(c) yes**, $\det = 5 \neq 0$.

Problem 5

In the plane, let $a = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

(a)

$$\det[a \ b] = \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = 1 \cdot 1 - 2 \cdot 3 = -5.$$

Geometrically $|-5| = 5$ is the area of the parallelogram spanned by a and b ; the sign records orientation (here b lies clockwise from a).

(b) The triangle $0, a, b$ is exactly half of that parallelogram, so its area is $\frac{5}{2}$.

(c) Three vectors u, v, w in $\mathbb{R}^2$ give a 2×3 system $Ac = 0$ with more unknowns than equations, hence a nontrivial solution: $w = \alpha u + \beta v$ for some α, β . In pictures, three vectors cannot all be “new directions” — the plane only has two. The determinant in (a) is the quantitative version of the same fact: as long as $\det \neq 0$, adding columns keeps independence possible; the moment a third column forces $\det = 0$, one of them became a combination of the others.

$\det = -5$, parallelogram area 5, triangle area $\frac{5}{2}$; any three vectors in $\mathbb{R}^2$ are dependent.

Problem 6

Let $W = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$

(a) With $p = (1, 1, 0)^T$, $q = (0, 1, 1)^T$, $r = (1, 0, 1)^T$ as columns,

$$\det[p \ q \ r] = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 1 \cdot (1 \cdot 1 - 0 \cdot 1) + 1 \cdot (1 \cdot 1 - 1 \cdot 0) = 1 + 1 = 2 \neq 0.$$

Three independent vectors in $\mathbb{R}^3$: they are a basis of their own span, so $\dim W = 3$.

(b) A 3-dimensional subspace of $\mathbb{R}^3$ is all of $\mathbb{R}^3$ — there is no room for anything bigger. So $W = \mathbb{R}^3$, and $\{p, q, r\}$ is a basis of $\mathbb{R}^3$.

(c) With $r' = (1, 0, -1)^T$,

$$\det[p \ q \ r'] = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} = 1 \cdot (1 \cdot (-1) - 0 \cdot 1) + 1 \cdot (1 \cdot 1 - 1 \cdot 0) = -1 + 1 = 0.$$

Now rank 2, so $\dim W' = 2$ — one dimension was lost. Which one? Solve $ap + bq = r'$:

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \Rightarrow a = 1, \ a + b = 0 \Rightarrow b = -1, \ b = -1 \checkmark$$

so $r' = p - q$.

(a) $\dim W = 3$, **basis** $\{p, q, r\}$; **(b)** $W = \mathbb{R}^3$; **(c)** $\dim W' = 2$ **with** $r' = p - q$.

Problem 7

In $\mathbb{R}^3$ with respect to the basis $B = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$:

(a) “Three vectors” is not a basis; independence is what has to be checked, and the answer is the same determinant as in Problem 6(a): $\det B = 2 \neq 0$. So B is a basis.

(b) Solve $Bc = v$:

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix} \Rightarrow c_1 + c_3 = 2, \ c_1 + c_2 = 3, \ c_2 + c_3 = 3.$$

Adding and subtracting: from the first and second, $c_2 = 3 - c_1$ and $c_3 = 2 - c_1$; the third gives $3 - c_1 + 2 - c_1 = 3$, so $2c_1 = 2$, $c_1 = 1$ and hence $c_2 = 2$, $c_3 = 1$.

(c) Multiply back:

$$1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+0+1 \\ 1+2+0 \\ 0+2+1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix} \checkmark$$

For a general vector, the same elimination with (x, y, z) in place of $(2, 3, 3)$ gives $c_1 + c_3 = x$, $c_1 + c_2 = y$, $c_2 + c_3 = z$; solving, $c_1 = \frac{x+y-z}{2}$, $c_2 = \frac{-x+y+z}{2}$, $c_3 = \frac{x-y+z}{2}$. In matrix form,

$$[v]_B = B^{-1}v, \quad B^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}.$$

A quick consistency check on that fraction: the integer matrix inside has determinant 4, so $\det B^{-1} = \frac{4}{2^3} = \frac{1}{2}$, which matches $\frac{1}{\det B} = \frac{1}{2}$. $\checkmark$

$$[v]_B = (1, 2, 1)^T; \quad B^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}.$$

Problem 8

Let B be the basis from Problem 7 and let $C = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$.

- (a) The standard-coordinate vector of any v can be written both ways: $v = B[v]_B = C[v]_C$. Therefore $[v]_C = C^{-1}B[v]_B$, so $S = C^{-1}B$. With

$$C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad C^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix},$$

$$S = C^{-1}B = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}.$$

- (b) $[w]_B$ comes from Problem 7's system: $c_1 + c_3 = 3$, $c_1 + c_2 = 1$, $c_2 + c_3 = 2$, which gives $c_1 = 1$, $c_2 = 0$, $c_3 = 2$. For C we solve $Cc = w$:

$$d_1 + d_3 = 3, \quad d_2 + d_3 = 1, \quad d_3 = 2 \implies d_1 = 1, \quad d_2 = -1, \quad d_3 = 2.$$

Now check:

$$S[w]_B = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = [w]_C \checkmark$$

- (c) S is the change of coordinates from B -coordinates to C -coordinates: it takes the coefficients in one basis and returns the coefficients of the same vector in another basis, without moving the vector itself.

$$S = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}; [w]_B = (1, 0, 2)^T, [w]_C = (1, -1, 2)^T, \text{ and } S[w]_B = [w]_C.$$

Problem 9

Identify every polynomial of degree ≤ 2 with its coefficient vector relative to $(1, x, x^2)$ — the polynomial $a + bx + cx^2$ becomes (a, b, c) . Let

In coefficient vectors: $p_1 = (1, 2, -1)$, $p_2 = (3, 1, 2)$, $p_3 = (4, 3, 1)$, $q = (5, -1, 4)$. Everything now behaves exactly as in $\mathbb{R}^3$.

- (a) $(1, 2, -1) + (3, 1, 2) = (4, 3, 1)$, so $p_1 + p_2 - p_3 = 0$: a nontrivial relation, hence dependent. Equivalently $\det[p_1 \ p_2 \ p_3] = 0$.
- (b) Solve $\alpha p_1 + \beta p_2 = q$ in coefficients:

$$\alpha + 3\beta = 5, \quad 2\alpha + \beta = -1, \quad -\alpha + 2\beta = 4.$$

From the first, $\alpha = 5 - 3\beta$; substitute into the second: $10 - 6\beta + \beta = -1$, so $5\beta = 11$, $\beta = \frac{11}{5}$, $\alpha = 5 - \frac{33}{5} = -\frac{8}{5}$. Check the third equation: $-\left(-\frac{8}{5}\right) + 2 \cdot \frac{11}{5} = \frac{8}{5} + \frac{22}{5} = \frac{30}{5} = 6 \neq 4$.

The system is inconsistent, so q is not in the span. (With three equations and only two unknowns, a single failed equation is enough — the same “a row of zeros with a nonzero entry” test as Problem 4.)

- (c) We want $\alpha p_1 + \beta p_2$ to vanish at $x = 1$. Evaluating at 1: $p_1(1) = 1 + 2 - 1 = 2$, $p_2(1) = 3 + 1 + 2 = 6$, so we need $2\alpha + 6\beta = 0$, i.e. $\alpha = -3\beta$. Take $\beta = 1$:

$$p = -3p_1 + p_2 = -3(1 + 2x - x^2) + (3 + x + 2x^2) = -3 - 6x + 3x^2 + 3 + x + 2x^2 = 5x^2 - 5x.$$

Check: $p(1) = 5 - 5 = 0$ ✓. And indeed $p = 5x(x - 1)$, so it vanishes at $x = 0$ as well.

(a) $p_3 = p_1 + p_2$; (b) no; (c) $5x^2 - 5x = 5x(x - 1)$, up to any nonzero multiple.

Problem 10

An experiment records a temperature T and a pressure P :

- (a) The five points lie on a line exactly when there are a, b with $aT_i + b = P_i$ for all five — i.e. when the 5×2 system is consistent. Stack the data:

$$M = \begin{bmatrix} 0 & 1 \\ 2 & 1 \\ 4 & 1 \\ 6 & 1 \\ 8 & 1 \end{bmatrix}, \quad \text{rank } [T \mid P] = \text{rank} \begin{bmatrix} 0 & 1 \\ 2 & 2 \\ 4 & 3 \\ 6 & 41/10 \\ 8 & 49/10 \end{bmatrix} = 2.$$

Rank 2 on the data columns means the two columns T and P are independent, which is exactly the statement “not all points share one line”. (If the points were collinear, P would be a multiple of T plus a multiple of the all-ones column, and the rank would drop to 1.)

- (b) Using the first three readings, $a \cdot 0 + b = 1$ gives $b = 1$; $2a + b = 2$ gives $a = \frac{1}{2}$; and $4a + b = 3$ confirms $3 = 3$ ✓.
- (c) Model $P = \frac{1}{2}T + 1$:

$$T = 6 : \quad P_{\text{model}} = 4.00, \text{ residual} = 4.10 - 4.00 = +0.10, \frac{0.10}{4.10} \approx +2.4\%; \quad T = 8 : \quad P_{\text{model}} = 5.00, \text{ residual} = 4.90 - 5.00 = -0.10, \frac{-0.10}{4.90} \approx -2.0\%$$

- (d) Each reading contributes one linear constraint ($aT_i + b = P_i$) on two unknowns (a, b). Three constraints on two unknowns must be redundant — some point is a combination of the others — so *any* three points are collinear. Five constraints on two unknowns are generically inconsistent, because $n + 1$ vectors in $\mathbb{R}^n$ are dependent (Problem 11) and a dependent system can be either inconsistent or redundant; here the data chose inconsistent. That is why experimenters *fit* lines instead of demanding exact ones.

(a) no — rank 2; (b) $P = \frac{1}{2}T + 1$; (c) $+0.10$ ($+2.4\%$) at $T = 6$, -0.10 (-2.0%) at $T = 8$; (d) more constraints than unknowns forces a dependence, and dependence means redundancy or contradiction — never a new direction.

Problem 11

Prove that a linearly independent set in $\mathbb{R}^n$ contains at most n vectors. Give two proofs:

- (a) Let A be the matrix whose k columns are the vectors. A has n rows and k columns, so its row echelon form has at most n pivots — one per row. The system $Ac = 0$ has a free variable for every non-pivot column, so it has a nontrivial solution unless every column is a pivot column. Nontrivial solution $\iff$ dependent, so an independent set forces every column to be a pivot column, hence $k \leq n$.
- (b) Same statement from the other side: A is $n \times k$ with $k > n$, and each of the n equations in the homogeneous system $Ac = 0$ can remove at most one degree of freedom, leaving at least one free variable. So there is a nonzero c with $Ac = 0$, i.e. $\sum c_i v_i = 0$ with not all c_i zero — dependence. If the vectors also happen to be n of them (a square A), this is exactly $\det A = 0$, which is Set 1's criterion: a singular matrix has a nontrivial null vector.

Equality is achievable: the standard vectors $e_1, \dots, e_n$ are n independent vectors in $\mathbb{R}^n$, so the bound is sharp.

At most n vectors, and n is attainable.

Problem 12

For a square matrix A , prove the chain of equivalences

Write the columns of A as $a_1, \dots, a_n$.

$(\det A \neq 0) \Rightarrow$ (independent): if the columns were dependent there would be $x \neq 0$ with $Ax = 0$; but $\det A = 0$ is precisely the failure of the elimination in Set 2 to reach a pivot in every row, so a free variable exists, so $Ax = 0$ has a nontrivial solution. Contrapositive gives the forward direction.

(independent) $\iff$ ($Ax = 0$ trivial): independence is *defined* as “no nontrivial combination of columns equals zero”, which is exactly the statement that $Ax = 0$ has no nontrivial solution — no theorem needed.

$(Ax = 0 \text{ trivial}) \iff (\text{invertible})$: Gaussian elimination writes $Ax = b$ as $Ux = \tilde{b}$ with U the echelon form of A . Every b has a unique solution exactly when U has a pivot in every row and no free variables — i.e. when the homogeneous case has only $x = 0$. Hence A is invertible exactly then.

So all four statements are one fact wearing four coats: *no free variables*.

Now the two matrices:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} : \quad \det A = 1 \cdot (1 \cdot 1 - 1 \cdot 0) - 2 \cdot (0 \cdot 1 - 1 \cdot 1) + 0 = 1 + 2 = 3 \neq 0.$$

All four conditions hold: columns independent, $Ax = 0$ only $x = 0$, and A invertible. (Its inverse is

$$\frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & -1 \\ -1 & 2 & 1 \end{bmatrix}, \text{ if you want it.})$$

$$A' = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{bmatrix} : \quad \text{row 2} = 2 \cdot \text{row 1, so the echelon form has a zero row and } \det A' = 0.$$

Equivalently, in terms of columns: $a_1 = (1, 2, 1)^T$, $a_2 = (2, 4, 0)^T$, $a_3 = (3, 6, 1)^T$, and $a_3 = a_1 + a_2$ — the columns are dependent. (Careful here: it is *not* $a_2 = 2a_1$; $2a_1$ would be $(2, 4, 2)^T$. Compare the third entry.)

Reduce $A'x = 0$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 - R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -2 & -2 \end{bmatrix},$$

$$\xrightarrow{R_3 \leftarrow R_3 / (-2)} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

$x_2 = -x_3$, and the first row gives $x_1 = -2x_2 - 3x_3 = 2x_3 - 3x_3 = -x_3$. Taking $x_3 = t$:

$$x = t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

So a basis of the solution set is $\{(-1, -1, 1)^T\}$, it is a line (dimension 1, matching $\text{rank } A' = 2 = 3 - 1$), and A' is not invertible.

A: all four conditions, $\det = 3$. A' : none of them, $\det = 0$, solution set = $\{t(-1, -1, 1)^T\}$.